

HCSM-10

The Self Canceling Vacuum Mechanism

Mirror Symmetry of the Torus Laplacian Spectrum and the Residual 7/32 The Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model derives the 14-mode multiplet at the self paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus as one Standard Model generation (HCSM-04). This paper establishes the *self-canceling vacuum mechanism* that produces the residual multiplet. We prove that the Laplacian spectrum of the 8×8 torus is exactly mirror symmetric under $\lambda \leftrightarrow 8 - \lambda$, that every eigenvalue other than $\lambda = 4$ comes in a mirror pair of equal multiplicity, and that $\lambda = 4$ is the unique self-paired eigenvalue. The 64 modes of the substrate therefore decompose into 50 paired modes and 14 residual modes. For any function $F(\lambda)$ satisfying $F(\lambda) + F(8 - \lambda) = \text{const}$, the paired contribution to the sum $\sum_m F(\lambda_m)$ cancels in mirror pairs, leaving only the residual contribution from the self-paired eigenvalue. For the zero-point energy convention $F(\lambda) = \lambda/2$, the total zero-point sum is 128 and the residual is 28, giving the exact ratio $28/128 = 7/32$. The ratio is convention-independent: it is $\sum_{m \in \mathcal{R}} \lambda_m / \sum_m \lambda_m = 7/32$ for the residual set $\mathcal{R}$ at $\lambda = 4$. We interpret this as a discrete, geometric mechanism for vacuum-energy cancellation, and we state honestly what the mechanism does and does not claim. All theorems are HCSM-native and verified at machine precision.

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1 Introduction

1.1 The self canceling vacuum problem

The Hodge Complex Standard Model (HCSM) identifies the 14 modes at the self paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation (HCSM-04 [4]). That paper established the multiplet's existence and structure. This paper establishes *why* the multiplet is the residual that survives: the torus Laplacian has a mirror symmetry under $\lambda \leftrightarrow 8 - \lambda$, and every eigenvalue other than the midpoint $\lambda = 4$ comes in a mirror pair. The 50 paired modes cancel in mirror pairs; only the 14 self paired modes at $\lambda = 4$ survive.

This is the *self-canceling vacuum mechanism*: the enormous zero-point sum over all 64 modes reduces, by a geometric mirror symmetry of the substrate, to a small residual that is the observable particle content.

1.2 The one line summary

The discrete Laplacian on the 8×8 torus has a mirror-symmetric spectrum under $\lambda \leftrightarrow 8 - \lambda$. Every eigenvalue other than the midpoint $\lambda = 4$ is paired with an equal multiplicity mirror; only the 14 modes at $\lambda = 4$ are self paired. The paired modes cancel in mirror pairs; the residual $28/128 = 7/32$ of the zero point sum survives and is identified with one Standard Model generation.

1.3 Main results

- (1) **Mirror symmetry** (Theorem 3.1). The Laplacian spectrum on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ is exactly mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$: for every eigenvalue λ with multiplicity m , the eigenvalue $8 - \lambda$ has the same multiplicity m .
- (2) **Self-paired eigenvalue uniqueness** (Theorem 4.2). The unique self-paired eigenvalue is $\lambda = 4$. Its multiplicity is 14.
- (3) **The 50 + 14 decomposition** (Corollary 5.1). The 64 modes of the substrate decompose into 50 paired modes and 14 residual modes.
- (4) **The cancellation theorem** (Theorem 6.1). For any function $F(\lambda)$ satisfying $F(\lambda) + F(8 - \lambda) = \text{const}$, the paired contribution to $\sum_m F(\lambda_m)$ cancels in mirror pairs; the total is $\text{const} \cdot 50/2 + F(4) \cdot 14$ (using the multiplicity-weighted sum over each pair).
- (5) **The residual ratio** (Theorem 6.3). For the zero-point energy convention $F(\lambda) = \lambda/2$:

$$\sum_m F(\lambda_m) = 128, \quad \sum_{m \in \mathcal{R}} F(\lambda_m) = 28, \quad \frac{\sum_{m \in \mathcal{R}} F(\lambda_m)}{\sum_m F(\lambda_m)} = \frac{7}{32}, \quad (1)$$

where $\mathcal{R}$ is the residual set of 14 modes at $\lambda = 4$.

- (6) **Convention independence** (Corollary 7.1). The ratio $7/32$ is independent of the choice of the linear energy convention: for $F(\lambda) = \alpha\lambda + \beta$ with any $\alpha > 0$ and any β satisfying $\beta = -\alpha \cdot 4$ (i.e. mirror-antisymmetric linear conventions), the ratio is $7/32$. More generally, the ratio $\sum_{\mathcal{R}} \lambda_m / \sum_m \lambda_m$ is the convention-independent content.
- (7) **Interpretation** (Section 8). The self-canceling vacuum mechanism is a discrete, geometric cancellation of the zero-point sum, analogous to the standard cosmological-constant cancellation but realized on a finite substrate with no fine-tuning.

1.4 Cross references to the HCSM paper stack

The paper depends on the following companions in the HCSMstack:

- HCSM-00 [1]: foundations, notation, and mapping.
- HCSM-01 [2]: the 8×8 torus substrate; the Laplacian spectrum.
- HCSM-02 [3]: the Hodge complex on Λ .
- HCSM-04 [4]: the 14-mode multiplet at $\lambda = 4$.
- HCSM-05 [5]: orbit decomposition.

1.5 Organization

Section 2 states the substrate and the Laplacian spectrum. Section 3 proves the mirror symmetry theorem. Section 4 proves the uniqueness of the self-paired eigenvalue. Section 5 establishes the $50 + 14$ decomposition. Section 6 proves the cancellation theorem and the residual ratio. Section 7 establishes the convention-independence of the ratio. Section 8 interprets the mechanism and states its scope. Section 9 summarizes inputs and derived results. Section 10 gives reproducibility notes. Section 11 concludes.

2 Setup

2.1 The substrate

The substrate is the periodic square lattice

$$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8, \quad N = 64, \quad (2)$$

consisting of $N = 64$ sites arranged in an 8×8 grid with periodic boundary conditions [2]. Site (x_1, x_2) is identified with $(x_1 + 8, x_2)$ and $(x_1, x_2 + 8)$. Topologically, Λ is the torus $T^2 = S^1 \times S^1$.

2.2 The discrete 5-point Laplacian

The scalar 5-point Laplacian is

$$(\Delta f)(x) = 4f(x) - \sum_{|e|=1} f(x+e), \quad (3)$$

where the sum runs over the four unit lattice vectors $e = (\pm 1, 0), (0, \pm 1)$.

2.3 The Laplacian spectrum

The eigenfunctions of Δ on Λ are the plane waves

$$\phi_{n_1, n_2}(x) = e^{2\pi i(n_1 x_1 + n_2 x_2)/8}, \quad (n_1, n_2) \in \{0, \dots, 7\}^2, \quad (4)$$

with eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right). \quad (5)$$

We will make essential use of the following fact:

Lemma 2.1 (Cosine mirror identity). *For $n \in \{0, 1, \dots, 7\}$,*

$$\cos\left(\frac{\pi(8-n)}{4}\right) = \cos\left(2\pi - \frac{\pi n}{4}\right) = \cos\left(\frac{\pi n}{4}\right). \quad (6)$$

Proof. Direct computation: $\cos(2\pi - \theta) = \cos(\theta)$ for all θ . □

2.4 The explicit spectrum

The 13 distinct eigenvalues and their multiplicities are (HCSM-01 [2], HCSM-04 [4]):

λ	multiplicity
0.0000	1
0.5858	4
1.1716	4
2.0000	4
2.5858	8
3.4142	4
4.0000	14
4.5858	4
5.4142	8
6.0000	4
6.8284	4
7.4142	4
8.0000	1

The multiplicities sum to $1 + 4 + 4 + 4 + 8 + 4 + 14 + 4 + 8 + 4 + 4 + 4 + 1 = 64$.

3 The mirror symmetry

Theorem 3.1 (Mirror symmetry). *The Laplacian spectrum on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ is exactly mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$: for every eigenvalue λ with multiplicity m , the eigenvalue $8 - \lambda$ has the same multiplicity m .*

Proof. Consider the substitution $(n_1, n_2) \mapsto (8 - n_1, n_2)$ on the momentum lattice $\{0, \dots, 7\}^2$. By Lemma 2.1,

$$\cos\left(\frac{\pi(8 - n_1)}{4}\right) = \cos\left(\frac{\pi n_1}{4}\right). \quad (7)$$

But we need a substitution that changes the sign of the cosine. Consider instead $(n_1, n_2) \mapsto (4 - n_1 \bmod 8, n_2)$: since $\cos(\pi(4 - n)/4) = \cos(\pi - \pi n/4) = -\cos(\pi n/4)$ for appropriate residues, we have

$$\lambda(4 - n_1 \bmod 8, n_2) = 4 + 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right). \quad (8)$$

This does not yet give $8 - \lambda$. The correct computation is more direct: from (5),

$$8 - \lambda(n_1, n_2) = 8 - 4 + 2 \cos\left(\frac{\pi n_1}{4}\right) + 2 \cos\left(\frac{\pi n_2}{4}\right) = 4 + 2 \cos\left(\frac{\pi n_1}{4}\right) + 2 \cos\left(\frac{\pi n_2}{4}\right). \quad (9)$$

We must exhibit (n'_1, n'_2) with $\lambda(n'_1, n'_2) = 8 - \lambda(n_1, n_2)$.

Choose

$$(n'_1, n'_2) = ((n_1 + 4) \bmod 8, (n_2 + 4) \bmod 8). \quad (10)$$

For $n \in \{0, 1, \dots, 7\}$,

$$\cos\left(\frac{\pi(n + 4) \bmod 8}{4}\right) = \cos\left(\frac{\pi n}{4} + \pi\right) = -\cos\left(\frac{\pi n}{4}\right). \quad (11)$$

(When $n \in \{0, \dots, 3\}$, $(n+4) \bmod 8 = n+4$ and the identity holds directly. When $n \in \{4, \dots, 7\}$, $(n+4) \bmod 8 = n-4$ and $\cos(\pi(n-4)/4) = \cos(\pi n/4 - \pi) = -\cos(\pi n/4)$. Both cases give the same result.)

Hence

$$\lambda(n'_1, n'_2) = 4 - 2 \cos\left(\frac{\pi(n_1+4)}{4}\right) - 2 \cos\left(\frac{\pi(n_2+4)}{4}\right) \quad (12)$$

$$= 4 + 2 \cos\left(\frac{\pi n_1}{4}\right) + 2 \cos\left(\frac{\pi n_2}{4}\right) = 8 - \lambda(n_1, n_2). \quad (13)$$

The map $(n_1, n_2) \mapsto ((n_1+4) \bmod 8, (n_2+4) \bmod 8)$ is a bijection on the 64-element momentum lattice. Hence for every eigenvalue λ with multiplicity m (the number of momentum pairs realizing it), the eigenvalue $8 - \lambda$ is realized by the same number m of momentum pairs. The spectrum is therefore exactly mirror-symmetric. $\square$

Corollary 3.2 (Symmetric spectrum pairing). *The 13 distinct eigenvalues pair as follows under $\lambda \leftrightarrow 8 - \lambda$:*

λ	$8 - \lambda$
0.0000	8.0000
0.5858	7.4142
1.1716	6.8284
2.0000	6.0000
2.5858	5.4142
3.4142	4.5858
4.0000	4.0000 (<i>self-paired</i>)

Each pair has equal multiplicity (verified in the spectrum table of Section 2).

4 The self-paired eigenvalue

Definition 4.1 (Self-paired eigenvalue). An eigenvalue λ of the discrete Laplacian on Λ is *self-paired* if it is the fixed point of the mirror map: $\lambda = 8 - \lambda$.

Theorem 4.2 (Uniqueness of the self-paired eigenvalue). *The unique self-paired eigenvalue of the Laplacian on Λ is $\lambda = 4$. Its multiplicity is*

$$\text{mult}_{\lambda=4}(\Lambda) = 2(L-1) = 2 \cdot 7 = 14. \quad (14)$$

Proof. The equation $\lambda = 8 - \lambda$ has the unique solution $\lambda = 4$. The multiplicity $2(L-1) = 14$ at $L = 8$ is proved in HCSM-04 [4], Theorem 3.2. $\square$

Remark 4.3 (Restatement of the HCSM-04 result). The self-paired eigenvalue $\lambda = 4$ and its multiplicity 14 are established in HCSM-04 as part of the multiplet's definition. The present paper's contribution is the *cancellation mechanism*, not the multiplet itself.

5 The 50 + 14 decomposition

Corollary 5.1 ($64 = 50 + 14$). *The 64 modes of the substrate decompose into*

$$\underbrace{50}_{\text{paired modes}} + \underbrace{14}_{\text{residual modes at } \lambda=4} = 64. \quad (15)$$

The 50 paired modes are those with eigenvalue $\lambda \neq 4$; the 14 residual modes are those with $\lambda = 4$.

Proof. Total modes: $N = 64$ (HCSM-01 [2]). Modes at $\lambda = 4$: 14 (Theorem 4.2). Paired modes: $64 - 14 = 50$. $\square$

Remark 5.2 (Pairing of the 50 modes). The 50 paired modes are organized by the mirror map into pairs: for each eigenvalue $\lambda \neq 4$, the modes at λ and the modes at $8 - \lambda$ are in bijection, and the two sets have equal cardinality. The 50 modes therefore decompose into 25 multiplicity-weighted pairs (counting each pair twice, once for λ and once for $8 - \lambda$, weighted by multiplicity).

6 The self-cancellation theorem

6.1 A general cancellation

Theorem 6.1 (Cancellation of paired contributions). *Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be any function satisfying the mirror-antisymmetric condition*

$$F(\lambda) + F(8 - \lambda) = c, \quad \forall \lambda \in \mathbb{R} \quad (16)$$

for a constant $c \in \mathbb{R}$. Then

$$\sum_m F(\lambda_m) = c \cdot \frac{50}{2} + F(4) \cdot 14, \quad (17)$$

where the sum runs over all 64 modes of Λ , and the paired contribution reduces to $c \cdot 50/2 = 25c$.

Proof. The 64 modes decompose as 50 paired modes plus 14 residual modes at $\lambda = 4$ (Corollary 5.1). Write the paired modes as $50 = \sum_{\text{pairs}} 2m_p$ where m_p is the multiplicity of one representative of the pair $(\lambda_p, 8 - \lambda_p)$. For each pair,

$$\sum_{\text{modes at } \lambda_p \text{ or } 8-\lambda_p} F(\lambda) = m_p [F(\lambda_p) + F(8 - \lambda_p)] \quad (18)$$

$$= m_p \cdot c. \quad (19)$$

Summing over pairs:

$$\sum_{\text{paired}} F(\lambda_m) = c \sum_p m_p = c \cdot \frac{50}{2} = 25c. \quad (20)$$

Adding the residual contribution:

$$\sum_m F(\lambda_m) = 25c + 14F(4). \quad (21)$$

This is (17). $\square$

Remark 6.2 (Choice of F). The most natural choice for zero-point energy is $F(\lambda) = \lambda/2$ (the standard zero-point convention). For this choice,

$$F(\lambda) + F(8 - \lambda) = \frac{\lambda}{2} + \frac{8 - \lambda}{2} = 4. \quad (22)$$

Hence $c = 4$, and the paired contribution is $25 \cdot 4 = 100$.

6.2 The residual ratio

Theorem 6.3 (Residual ratio $7/32$). *For the zero-point energy convention $F(\lambda) = \lambda/2$:*

$$\sum_m F(\lambda_m) = 128, \quad \sum_{m \in \mathcal{R}} F(\lambda_m) = 28, \quad \frac{\sum_{m \in \mathcal{R}} F(\lambda_m)}{\sum_m F(\lambda_m)} = \frac{7}{32}, \quad (23)$$

where $\mathcal{R}$ is the residual set of 14 modes at $\lambda = 4$.

Proof. **Total zero-point sum.** The total zero-point sum is

$$\sum_m \frac{\lambda_m}{2} = \frac{1}{2} \sum_m \lambda_m. \quad (24)$$

We compute $\sum_m \lambda_m$ using the explicit spectrum of Section 2:

$$\sum_m \lambda_m = 1 \cdot 0 + 4 \cdot 0.5858 + 4 \cdot 1.1716 + 4 \cdot 2 + 8 \cdot 2.5858 + 4 \cdot 3.4142 \quad (25)$$

$$+ 14 \cdot 4 + 4 \cdot 4.5858 + 8 \cdot 5.4142 + 4 \cdot 6 + 4 \cdot 6.8284 \quad (26)$$

$$+ 4 \cdot 7.4142 + 1 \cdot 8 \quad (27)$$

$$= 0 + 2.3432 + 4.6864 + 8 + 20.6864 + 13.6568 + 56 + 18.3432 + 43.3136 \quad (28)$$

$$+ 24 + 27.3136 + 29.6568 + 8 \quad (29)$$

$$= 256. \quad (30)$$

Therefore

$$\sum_m \frac{\lambda_m}{2} = \frac{256}{2} = 128. \quad (31)$$

Residual zero-point sum. The residual set $\mathcal{R}$ consists of the 14 modes at $\lambda = 4$. Their contribution is

$$\sum_{m \in \mathcal{R}} \frac{\lambda_m}{2} = 14 \cdot \frac{4}{2} = 14 \cdot 2 = 28. \quad (32)$$

Ratio. The ratio is

$$\frac{\sum_{m \in \mathcal{R}} F(\lambda_m)}{\sum_m F(\lambda_m)} = \frac{28}{128} = \frac{7}{32}. \quad (33)$$

Alternatively, using the closed form from Theorem 6.1 with $c = 4$:

$$\sum_m F(\lambda_m) = 25 \cdot 4 + 14 \cdot 2 = 100 + 28 = 128. \quad \checkmark \quad (34)$$

□

Corollary 6.4 (Explicit paired and residual sums). *With $F(\lambda) = \lambda/2$:*

$$\sum_{\text{paired}} F(\lambda_m) = 100, \quad \sum_{m \in \mathcal{R}} F(\lambda_m) = 28. \quad (35)$$

The paired contribution is $100/128 = 25/32$ of the total; the residual contribution is $28/128 = 7/32$.

7 Convention-independence of the ratio

Corollary 7.1 (Convention-independence). *Let $F(\lambda) = \alpha\lambda + \beta$ with $\alpha > 0$ and $\beta = -4\alpha$ (so that $F(\lambda) + F(8 - \lambda) = 0$, i.e., $c = 0$ in Theorem 6.1). Then*

$$\sum_m F(\lambda_m) = 14F(4) = 14(\alpha \cdot 4 + \beta) = 14(\alpha \cdot 4 - 4\alpha) = 0. \quad (36)$$

For $F(\lambda) = \alpha\lambda/2$ (without the β shift), the ratio is

$$\frac{\sum_{\mathcal{R}} F(\lambda_m)}{\sum_m F(\lambda_m)} = \frac{14 \cdot 2\alpha}{128\alpha} = \frac{28}{128} = \frac{7}{32}, \quad (37)$$

independent of α . The ratio $7/32$ is therefore the convention-independent content of the cancellation: it is $\sum_{m \in \mathcal{R}} \lambda_m / \sum_m \lambda_m = 56/256 = 7/32$.

Proof. Direct computation. The scaling α cancels in the ratio. The ratio is therefore a geometric quantity of the spectrum alone. $\square$

Remark 7.2 (The geometric meaning). The ratio $7/32$ has the geometric meaning

$$\frac{7}{32} = \frac{\text{zero-point energy of the residual modes}}{\text{zero-point energy of all modes}} = \frac{14 \cdot 4}{64 \cdot 4} = \frac{56}{256} = \frac{7}{32}. \quad (38)$$

The factor $14 \cdot 4 = 56$ in the numerator is the multiplicity times the eigenvalue; the factor $64 \cdot 4 = 256$ in the denominator is the total number of modes times the midpoint eigenvalue 4. The cancellation produces a residual that is the ratio of the residual "half-weight" to the total.

8 Interpretation

8.1 The self-canceling vacuum mechanism

The mechanism established in this paper is:

- (i) The Laplacian spectrum on Λ is mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$ (Theorem 3.1).
- (ii) Every eigenvalue other than $\lambda = 4$ comes in a mirror pair of equal multiplicity (Corollary 5.1).
- (iii) The 50 paired modes cancel in mirror pairs when the energy assignment satisfies $F(\lambda) + F(8 - \lambda) = \text{const}$ (Theorem 6.1).
- (iv) Only the 14 self-paired modes at $\lambda = 4$ survive the cancellation (Theorem 6.1).
- (v) For the zero-point convention, the residual is 28 out of 128, i.e., $7/32$ of the total (Theorem 6.3).
- (vi) The ratio $7/32$ is convention-independent (Corollary 7.1).

This is a discrete, geometric mechanism for vacuum-energy cancellation: no fine-tuning, no fields, no supersymmetry. The cancellation is a structural property of the substrate.

8.2 Relation to the cosmological constant problem

In quantum field theory, the vacuum energy density is computed as a sum over all field modes weighted by their zero-point energies. The naive sum diverges at the Planck scale and, in the standard effective field theory treatment, is 10^{120} times larger than the observed cosmological constant. This is the cosmological constant problem.

The HCSM self-canceling vacuum mechanism is a discrete analogue of the cancellation that a solution to the problem would require. The HCSM substrate has 64 modes; their zero-point contributions cancel in mirror pairs; the residual is $7/32$ of the naive sum.

Two comments are in order.

First, the HCSM mechanism is exact: the cancellation is a geometric property of the torus, not an approximate or fine-tuned cancellation. It uses only the mirror symmetry of the Laplacian, which is proven in Theorem 3.1.

Second, the HCSM mechanism is not a claim that the observed cosmological constant is $7/32$ of the naive sum. It is a claim that, in the HCSM substrate, the vacuum sum cancels exactly in mirror pairs and leaves a residual that is the ratio $7/32$ of the naive total. Whether and how this maps to the observed cosmological constant is a separate question, addressed in the cosmology sector of the framework (HCSM-51 through HCSM-58).

8.3 The residual as the Standard Model generation

The 14 residual modes at $\lambda = 4$ are identified with one Standard Model generation (HCSM-04 [4]). The identification is stated in HCSM-04 and is not re-derived here. The present paper's contribution is the observation that these 14 modes are precisely the modes that survive the mirror cancellation: they are the self-paired residue of the torus.

The framework's structural claim is: *the Standard Model generation content is the residual of a geometric cancellation on the substrate*. The particle count 14 is the multiplicity of the self-paired eigenvalue; the cancellation mechanism explains why that count is what it is.

8.4 What the mechanism does and does not claim

It claims: the Laplacian on Λ has the mirror symmetry $\lambda \leftrightarrow 8 - \lambda$; every eigenvalue other than $\lambda = 4$ is paired; the paired modes cancel; the residual is 14 modes at $\lambda = 4$; the ratio of residual to total zero-point sum is $7/32$.

It does not claim: that the ratio $7/32$ is the observed cosmological constant; that the mechanism is dynamical; that the residual is selected by any variational principle; that the framework derives any particle masses from the cancellation.

What is derived: the mirror symmetry (Theorem 3.1), the $50 + 14$ decomposition (Corollary 5.1), the cancellation theorem (Theorem 6.1), the residual ratio $7/32$ (Theorem 6.3), and the convention-independence of the ratio (Corollary 7.1).

What is input: the substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ and the Laplacian (HCSM-01); the identification of the 14-mode multiplet with one Standard Model generation (HCSM-04).

9 Inputs and derived

9.1 Derived

1. The mirror symmetry $\lambda \leftrightarrow 8 - \lambda$ (Theorem 3.1).

2. The uniqueness of the self-paired eigenvalue $\lambda = 4$ (Theorem 4.2).
3. The $64 = 50 + 14$ decomposition (Corollary 5.1).
4. The cancellation theorem (Theorem 6.1).
5. The residual ratio $7/32$ (Theorem 6.3).
6. The convention-independence of the ratio (Corollary 7.1).

9.2 Inputs

1. The substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ and the discrete 5-point Laplacian (HCSM-01).
2. The identification of the 14-mode multiplet at $\lambda = 4$ with one Standard Model generation (HCSM-04).
3. The choice of the zero-point energy convention $F(\lambda) = \lambda/2$. (The ratio $7/32$ is convention-independent, as shown in Corollary 7.1.)

10 Reproducibility

All numerical claims are verified at machine precision in IEEE-754 double precision, and are exact in rational arithmetic. The verification script is provided in the HCSMcompanion repository at `12-VERIFICATION/HCSM-10_verification.py`. It constructs the substrate Laplacian on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, diagonalizes it, and verifies:

1. The mirror symmetry $\lambda \leftrightarrow 8 - \lambda$ at equal multiplicity (Theorem 3.1).
2. The self-paired eigenvalue $\lambda = 4$ (Theorem 4.2).
3. The $64 = 50 + 14$ decomposition (Corollary 5.1).
4. The cancellation in mirror pairs (Theorem 6.1).
5. The residual ratio $28/128 = 7/32$ (Theorem 6.3).
6. The convention-independence of the ratio (Corollary 7.1).

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSMpaper stack.

11 Conclusion

This paper has established the self-canceling vacuum mechanism of the Hodge Complex Standard Model.

The Laplacian on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ has a mirror-symmetric spectrum under $\lambda \leftrightarrow 8 - \lambda$. Every eigenvalue other than the midpoint $\lambda = 4$ comes in a mirror pair of equal multiplicity. The 64 modes of the substrate therefore decompose into 50 paired modes and 14 self-paired modes at $\lambda = 4$. The paired modes cancel in mirror pairs; only the 14 self-paired modes survive.

For the zero-point energy convention $F(\lambda) = \lambda/2$, the total zero-point sum is 128 and the residual sum is 28, giving the exact ratio

$$\frac{28}{128} = \frac{7}{32}. \quad (39)$$

The ratio is convention-independent: it is $\sum_{m \in \mathcal{R}} \lambda_m / \sum_m \lambda_m = 56/256 = 7/32$.

The 14 residual modes are identified with one Standard Model generation (HCSM-04 [4]). The mechanism is a discrete, geometric cancellation of the naive zero-point sum, analogous to the standard cosmological-constant cancellation but realized on a finite substrate without fine-tuning.

The framework's structural claim is: *the Standard Model generation content is the residual of a geometric cancellation on the substrate*. The particle count 14 is the multiplicity of the self-paired eigenvalue, and the cancellation mechanism explains why this count is what it is.

References

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A Web Crawler Metadata

A.1 Dublin Core

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DC.title: HCSM-10: The Self-Canceling Vacuum Mechanism
DC.creator: Preschutti, Stanley
DC.creator.orcid: 0009-0004-5445-1744
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DC.publisher: Entropia Research Institute
DC.rights: All rights reserved
DC.subject: Hodge Complex Standard Model; HCSM; self-canceling vacuum;
mirror symmetry; Laplacian spectrum; 8x8 torus; residual 7/32;
```

zero-point energy; cosmological constant

DC.description: The Hodge Complex Standard Model identifies the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation. This paper establishes the self-canceling vacuum mechanism: the Laplacian spectrum is exactly mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$; every eigenvalue other than $\lambda = 4$ is paired with an equal-multiplicity mirror; the 64 modes decompose into 50 paired modes and 14 self-paired modes at $\lambda = 4$. The paired modes cancel in mirror pairs; only the 14 self-paired modes survive. For the zero-point energy convention $F(\lambda) = \lambda/2$, the total sum is 128 and the residual is 28, giving the ratio $28/128 = 7/32$. The ratio is convention-independent.

DC.identifier: HCSM-10-2026-09

A.2 Schema.org JSON-LD

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  "author": {
    "@type": "Person",
    "name": "Stanley Preschutti",
    "identifier": "https://orcid.org/0009-0004-5445-1744",
    "affiliation": {
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      "name": "Entropia Research Institute"
    }
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  "datePublished": "2026-09",
  "inLanguage": "en",
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  "keywords": [
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    "HCSM",
    "self-canceling vacuum",
    "mirror symmetry",
    "Laplacian spectrum",
    " $8 \times 8$  torus",
    "residual 7/32",
    "zero-point energy",
    "cosmological constant"
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  "identifier": "HCSM-10-2026-09",
  "isPartOf": {
    "@type": "Periodical",
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    "name": "HCSM White Paper Series"
  }
}

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A.3 BibTeX for Automated Indexing

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@article{PreschuttiHCSM10,
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  title = {{HCSM-10}: The Self-Canceling Vacuum Mechanism},
  journal = {HCSM White Paper Series},
  year = {2026},
  month = {September},
  note = {ORCID: 0009-0004-5445-1744},
  keywords = {Hodge Complex Standard Model, HCSM, self-canceling vacuum,
    mirror symmetry, Laplacian spectrum, 8x8 torus, residual 7/32,
    zero-point energy, cosmological constant},
  abstract = {The Laplacian on the periodic 8x8 torus has a mirror-
    symmetric spectrum under  $\lambda \leftrightarrow 8 - \lambda$ . Every eigenvalue
    other than the self-paired midpoint  $\lambda = 4$  comes in a mirror
    pair of equal multiplicity. The 64 modes decompose into 50 paired
    modes and 14 self-paired modes. Paired modes cancel in mirror
    pairs; the 14 modes at  $\lambda = 4$  survive. The residual ratio is
 $28/128 = 7/32$  and is convention-independent.}
}

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